Φ-Singular and Φ-Cosingular Operators: Verification, Literature Status, and New Reductions

Research report prepared from the supplied solver packet

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Abstract

The four main arguments in the supplied packet are mathematically sound: reflexive adjoint duality becomes an equivalence; one-sided square embeddings/quotients identify $\Phi\mathcal{S}$ and $\Phi\mathcal{C}$ with the corresponding perturbation classes; subprojective and superprojective operators cannot be strict witnesses; and the stated extension/lifting obstructions are valid. Two editorial corrections are needed: the supporting paper's relevant result is Corollary 2.3, not Corollary 2.4, and the corollary comparing strictness under adjoints must retain its stated extra hypothesis because strict singularity and strict cosingularity are not mutually dual in full generality.

A literature search through 22 June 2026 found no proof or counterexample settling $\mathcal{SS} = \Phi\mathcal{S}$ or $\mathcal{SC} = \Phi\mathcal{C}$, and no later resolution of the associated subspace questions. The report records several further rigorous reductions. Globally, the operator-ideal question is exactly equivalent to the equality question. One half of each same-component multiplication problem is automatic, and linearity would imply the other half. Any known perturbation-class witness outside $\Phi\mathcal{S}$ must have an offending upper semi-Fredholm column with uncomplemented range; dually, an offending lower semi-Fredholm row must have uncomplemented kernel. Finally, a new quotient-algebra criterion is proved: a Banach space with $\mathcal{L}(X)/\mathcal{SS}(X) \cong \mathbb{F}[\varepsilon]/(\varepsilon^n)$, $n \geq 2$, would immediately produce a strict operator in $\Phi\mathcal{S}(X) \setminus \mathcal{SS}(X)$, with a dual statement for $\Phi\mathcal{C}$. No construction realizing this last quotient by strictly singular (or strictly cosingular) operators was found. Thus the global problem remains open, but the remaining bottleneck is made substantially more concrete.

1 The problem and notation

Let X, Y be Banach spaces. For $S, K \in \mathcal{L}(X,Y)$, put

$$(S,K)x = (Sx, Kx) \colon X \longrightarrow Y \oplus Y, \qquad [S,K](x_1,x_2) = Sx_1 + Kx_2 \colon X \oplus X \longrightarrow Y.$$

An operator is upper semi-Fredholm, written $T \in \Phi_+$, when it has closed range and finite-dimensional kernel; it is lower semi-Fredholm, written $T \in \Phi_-$, when its range is closed and finite-codimensional. Whenever the relevant semi-Fredholm component is nonempty, define

$$K \in \Phi\mathcal{S}(X,Y) \iff (S,K) \in \Phi_+ \Longrightarrow S \in \Phi_+ \quad (S \in \mathcal{L}(X,Y)),$$
$$K \in \Phi\mathcal{C}(X,Y) \iff [S,K] \in \Phi_- \Longrightarrow S \in \Phi_- \quad (S \in \mathcal{L}(X,Y)).$$

Aiena–González–Martínez-Abejón identified these with Friedman's conditions (C) and (D), respectively, and proved

$$\mathcal{SS}(X,Y) \subseteq \Phi\mathcal{S}(X,Y) \subseteq P\Phi_+(X,Y), \qquad \mathcal{SC}(X,Y) \subseteq \Phi\mathcal{C}(X,Y) \subseteq P\Phi_-(X,Y).$$

González–Salas-Brown asked whether the first inclusions can be strict, whether the intermediate classes are linear subspaces or operator ideals, and whether they satisfy the same-component multiplication rules enjoyed by the perturbation classes [1, 7].

Literature status. The 2012 paper explicitly left open whether conditions (C) and (D) characterize strictly singular and strictly cosingular operators. The 2023 paper restated the problems for $\Phi\mathcal{S}$ and $\Phi\mathcal{C}$. Searches by the exact terminology, by the older condition $(C)/(D)$ terminology, through citation trails, and through later work of the authors found no subsequent full proof or counterexample up to 22 June 2026. Accordingly, this report does not claim to settle the global equality.

2 Audit of the supplied packet

This section states the four principal assertions in a corrected form and gives enough of each proof to make the verification independent.

2.1 Reflexive adjoint duality

Proposition 2.1 (verified)**.** *Let X, Y be reflexive. Whenever the displayed components are defined, for every $K \in \mathcal{L}(X,Y)$,*

$$K \in \Phi\mathcal{C}(X,Y) \iff K^* \in \Phi\mathcal{S}(Y^*,X^*), \qquad K \in \Phi\mathcal{S}(X,Y) \iff K^* \in \Phi\mathcal{C}(Y^*,X^*).$$

Proof. González–Salas-Brown proved the implications from the adjoint class to the original class. For the converse, take, for example, $K \in \Phi\mathcal{C}(X,Y)$, and suppose $(A, K^*) \in \Phi_+(Y^*, X^* \oplus X^*)$. Reflexivity implies that every $A\colon Y^* \to X^*$ has a preadjoint: under the canonical identifications there is $S\colon X \to Y$ with $A = S^*$. Since $[S,K]^* = (S^*,K^*)$, semi-Fredholm duality gives $[S,K] \in \Phi_-$. Hence $S \in \Phi_-$, so $A = S^* \in \Phi_+$. Thus $K^* \in \Phi\mathcal{S}$. The other converse uses $(S,K)^* = [S^*,K^*]$ in exactly the same way. $\square$

Remark 2.2. *The associated statement about strictness under adjoints must be qualified. In general $T^* \in \mathcal{SS} \Rightarrow T \in \mathcal{SC}$ and $T^* \in \mathcal{SC} \Rightarrow T \in \mathcal{SS}$, but the converses fail on arbitrary Banach spaces. Therefore the packet is correct only with its final qualification that the usual strictly singular/strictly cosingular adjoint duality must hold in the spaces at hand.*

2.2 One-sided square hypotheses

Proposition 2.3 (verified)**.** *Assume the relevant components are nonempty.*

(i) *If $Y \oplus Y$ is isomorphic to a closed subspace of Y, then $\Phi\mathcal{S}(X,Y) = P\Phi_+(X,Y)$.*

(ii) *If $X \oplus X$ is isomorphic to a quotient of X, then $\Phi\mathcal{C}(X,Y) = P\Phi_-(X,Y)$.*

Proof. For (i), let $U\colon Y \oplus Y \to Y$ be an into isomorphism and write $U(y_1,y_2) = Vy_1 + Wy_2$. If $K \in P\Phi_+(X,Y)$ and $(S,K) \in \Phi_+$, then $VS + WK = U(S,K) \in \Phi_+$. Proposition 2.2 of [7] gives $WK \in P\Phi_+$, so perturbing by $-WK$ yields $VS \in \Phi_+$. Since V is an into isomorphism, this is equivalent to $S \in \Phi_+$.

For (ii), let $Q\colon X \to X \oplus X$ be onto and write $Qx = (U_1x, U_2x)$. Then U_1 is onto. If $K \in P\Phi_-$ and $[S,K] \in \Phi_-$, then $[S,K]Q = SU_1 + KU_2 \in \Phi_-$. Subtracting $KU_2 \in P\Phi_-$ gives $SU_1 \in \Phi_-$. Since U_1 is onto, SU_1 and S have exactly the same range, proving $S \in \Phi_-$. $\square$

This is a genuine, though modest, sharpening of the isolated square-isomorphism hypotheses in [7]: an embedding is enough on the $\Phi\mathcal{S}$ side and a quotient is enough on the $\Phi\mathcal{C}$ side.

2.3 Subprojective and superprojective barriers

Recall that $K\colon X \to Y$ is subprojective if every infinite-dimensional subspace on which K is an isomorphism contains a further infinite-dimensional N for which $K(N)$ is complemented in Y. It is superprojective if every infinite-codimensional $M \subset Y$ for which $Q_M K$ is onto is contained in an infinite-codimensional N for which $K^{-1}(N)$ is complemented in X [5].

Proposition 2.4 (verified)**.** (i) *If $\Phi_+(X,Y) \neq \varnothing$ and K is subprojective, then*

$$K \in P\Phi_+(X,Y) \iff K \in \Phi\mathcal{S}(X,Y) \iff K \in \mathcal{SS}(X,Y).$$

(ii) *If $\Phi_-(X,Y) \neq \varnothing$ and K is superprojective, then*

$$K \in P\Phi_-(X,Y) \iff K \in \Phi\mathcal{C}(X,Y) \iff K \in \mathcal{SC}(X,Y).$$

Proof. Only the outer implication in each line is new. Suppose that a subprojective $K \in P\Phi_+$ is not strictly singular. On some infinite-dimensional $N \subset X$, $K|_N$ is an isomorphism and $K(N)$ is complemented. If $P\colon Y \to K(N)$ is a projection, $R = (K|_N)^{-1}$, and $A \in \Phi_+(X,Y)$, set $L = ARP \in \mathcal{L}(Y)$. Then $(A - LK)|_N = 0$, so $A - LK \notin \Phi_+$. But $LK \in P\Phi_+$, contradicting perturbation stability.

Dually, if a superprojective $K \in P\Phi_-$ is not strictly cosingular, choose infinite-codimensional $N \subset Y$ so that $Q_N K$ is onto and $E = K^{-1}(N)$ is complemented. The quotient $Q_N K$ then has a bounded right inverse R. Given $B \in \Phi_-(X,Y)$, put $T = RQ_N B$. The range of $B - KT$ is contained in N, so $B - KT \notin \Phi_-$, again contradicting $KT \in P\Phi_-$. □

Remark 2.5 (citation correction)**.** *The packet cites Corollary 2.4 of [5] for the statement that every map into a subprojective space is a subprojective operator, and every map out of a superprojective space is a superprojective operator. The correct citation is Corollary 2.3.*

Combining the proposition with the source paper gives a symmetric useful consequence: if either X or Y is subprojective, then $\mathcal{SS}(X,Y) = \Phi\mathcal{S}(X,Y)$; if either X or Y is superprojective, then $\mathcal{SC}(X,Y) = \Phi\mathcal{C}(X,Y)$, whenever the corresponding component is defined.

2.4 Extension and lifting obstructions

Proposition 2.6 (verified)**.** *Assume $\Phi_+(X,Y) \neq \varnothing$. If $K \in P\Phi_+(X,Y)$ is an isomorphism on an infinite-dimensional closed $M \subset X$, then for every $A \in \Phi_+(X,Y)$ the map*

$$A|_M (K|_M)^{-1}\colon K(M) \longrightarrow Y$$

has no bounded extension to all of Y. Consequently, if every bounded map from a closed subspace of Y into Y extends to an operator on Y, then

$$\mathcal{SS}(X,Y) = \Phi\mathcal{S}(X,Y) = P\Phi_+(X,Y).$$

Dually, assume $\Phi_-(X,Y) \neq \varnothing$. If $K \in P\Phi_-(X,Y)$ and $Q_M K\colon X \to Y/M$ is onto for an infinite-codimensional $M \subset Y$, then for every $B \in \Phi_-(X,Y)$, the map $Q_M B$ does not lift through $Q_M K$. Thus, if X has the stated lifting property for arbitrary quotient maps, then

$$\mathcal{SC}(X,Y) = \Phi\mathcal{C}(X,Y) = P\Phi_-(X,Y).$$

Proof. If the first transfer map extended to $L \in \mathcal{L}(Y)$, then $A - LK$ would vanish on M, while perturbation stability would put $A - LK$ in Φ_+, a contradiction. For the dual statement, a lift T satisfying $Q_M KT = Q_M B$ would force $\operatorname{ran}(B - KT) \subset M$, while perturbation stability would put $B - KT$ in Φ_-. The two equality conclusions follow from the standard inclusion chains. □

The proof is correct as written. The extension property is very strong, so it is best stated self-containedly rather than tied to literature on smooth nonlinear extensions.

3 The operator-ideal question is exactly the equality question

The following observation is already implicit in the 2012 maximality theorem, but it sharply organizes one of the source paper's questions.

Theorem 3.1 (maximality reduction)**.** *Under the natural convention that the classes are considered on all components where the semi-Fredholm class is nonempty:*

(i) *$\Phi\mathcal{S}$ is an operator ideal if and only if $\Phi\mathcal{S} = \mathcal{SS}$ componentwise.*

(ii) *$\Phi\mathcal{C}$ is an operator ideal if and only if $\Phi\mathcal{C} = \mathcal{SC}$ componentwise.*

Proof. Aiena–González–Martínez-Abejón proved that $\mathcal{SS}$ is the largest operator ideal whose defined components are contained in $P\Phi_+$, and that $\mathcal{SC}$ is the largest operator ideal whose defined components are contained in $P\Phi_-$ [1, Propositions 3.5 and 3.8]. If $\Phi\mathcal{S}$ were an operator ideal, the inclusion $\Phi\mathcal{S} \subseteq P\Phi_+$ would therefore imply $\Phi\mathcal{S} \subseteq \mathcal{SS}$, while the reverse inclusion is always true. The converse is immediate because $\mathcal{SS}$ is an operator ideal. The $\Phi\mathcal{C}$ proof is identical. □

Thus a positive answer to the global operator-ideal question would itself settle the strictness problem positively; conversely, any strict witness automatically proves that the corresponding class is not an operator ideal.

4 Linearity would force the full same-component multiplication rule

The source asks whether Proposition 2.2 for perturbation classes remains valid for $\Phi\mathcal{S}$ and $\Phi\mathcal{C}$. Half of each assertion does not require any extra assumption.

Proposition 4.1 (New reduction: unconditional half of multiplication)**.** *Let the indicated components be defined.*

(i) *If $K \in \Phi\mathcal{S}(X, Y)$ and $L \in \mathcal{L}(Y)$, then $LK \in \Phi\mathcal{S}(X, Y)$. If $U \in \mathcal{L}(X)$ is an automorphism, then $KU \in \Phi\mathcal{S}(X, Y)$.*

(ii) *If $K \in \Phi\mathcal{C}(X, Y)$ and $T \in \mathcal{L}(X)$, then $KT \in \Phi\mathcal{C}(X, Y)$. If $V \in \mathcal{L}(Y)$ is an automorphism, then $VK \in \Phi\mathcal{C}(X, Y)$.*

Proof. Suppose $(S, LK) \in \Phi_+$. With $D(y_1, y_2) = (y_1, Ly_2)$, we have $D(S, K) = (S, LK)$. Whenever DT is upper semi-Fredholm, T is upper semi-Fredholm: on a finite-codimensional subspace where DT is bounded below, T is bounded below as well. Hence $(S, K) \in \Phi_+$, and $S \in \Phi_+$. This proves $LK \in \Phi\mathcal{S}$. If $(S, KU) \in \Phi_+$, precomposition by U^{-1} gives $(SU^{-1}, K) \in \Phi_+$, so SU^{-1}, and hence S, is upper semi-Fredholm.

Now suppose $[S, KT] \in \Phi_-$. Its range $\operatorname{ran} S + \operatorname{ran}(KT)$ is a closed finite-codimensional subspace contained in $\operatorname{ran} S + \operatorname{ran} K$. Any linear subspace containing a closed finite-codimensional subspace is itself closed and finite-codimensional. Thus $[S, K] \in \Phi_-$, and $S \in \Phi_-$. This proves $KT \in \Phi\mathcal{C}$. The automorphism assertion follows by postcomposing $[S, VK]$ with V^{-1}. □

Corollary 4.2 (New reduction)**.** *If $\Phi\mathcal{S}(X,Y)$ is a linear subspace, then it satisfies both same-component multiplication rules*

$$LK,\ KT \in \Phi\mathcal{S}(X,Y) \quad (L \in \mathcal{L}(Y),\ T \in \mathcal{L}(X)).$$

If $\Phi\mathcal{C}(X,Y)$ is a linear subspace, it satisfies the analogous two rules.

Proof. Every element T of a unital Banach algebra is a sum of two invertibles: choose $\lambda > \|T\|$ and write $T = (T+\lambda I) + (-\lambda I)$. Apply the automorphism parts of Proposition 4.1 and then linearity. The multiplication on the other side is already unconditional. □

Consequently, on a fixed component the unresolved multiplication question is no harder than the unresolved linearity question.

5 Where a known counterexample must hide: uncomplemented geometry

Both perturbation classes are contained in the inessential operators $\mathcal{I}\backslash$ [1]. This gives a useful obstruction not visible in the bare definition.

Proposition 5.1 (New reduction: complemented regularizer obstruction)**.** (i) *Let $K \in P\Phi_+(X,Y)$. If $(S,K) \in \Phi_+(X, Y\oplus Y)$ and $\operatorname{ran}(S,K)$ is complemented in $Y \oplus Y$, then $S \in \Phi_+(X,Y)$.*

(ii) *Let $K \in P\Phi_-(X,Y)$. If $[S,K] \in \Phi_-(X\oplus X, Y)$ and $\ker[S,K]$ is complemented in $X \oplus X$, then $S \in \Phi_-(X,Y)$.*

Proof. For (i), complemented range and finite-dimensional kernel give a left regularizer $R\colon Y\oplus Y \to X$ with $R(S,K) = I_X - P$, where P has finite rank. Write $R = (A,B)$. Then

$$AS = I_X - (BK+P).$$

Because K is inessential and $\mathcal{I}\backslash$ is an operator ideal, $BK + P \in \mathcal{I}\backslash(X)$, so AS is Fredholm. If a composition AS is upper semi-Fredholm, then S is upper semi-Fredholm.

For (ii), complemented kernel and finite-codimensional range give a right regularizer $R\colon Y \to X\oplus X$ with $[S,K]R = I_Y - Q$, Q finite rank. Write $Ry = (Ay, By)$. Then

$$SA = I_Y - (KB+Q),$$

which is Fredholm. Hence $\operatorname{ran}(S)$, containing the finite-codimensional closed range of SA, is itself closed and finite-codimensional. □

Corollary 5.2 (localization of all known failures)**.** *If $K \in P\Phi_+(X,Y)\setminus\Phi\mathcal{S}(X,Y)$, every witnessing $S \notin \Phi_+$ with $(S,K)\in\Phi_+$ has uncomplemented column range. Dually, if $K \in P\Phi_-(X,Y)\setminus\Phi\mathcal{C}(X,Y)$, every witnessing lower semi-Fredholm row has uncomplemented kernel.*

This explains why the established counterexamples to $\mathcal{SS} = P\Phi_+$ and $\mathcal{SC} = P\Phi_-$ do not automatically answer the present question: their failure is concentrated in precisely the uncomplemented part of semi-Fredholm geometry.

6 A concrete route to a strict counterexample

The next criterion is algebraic and appears to be new. It does not by itself construct the required Banach space, but it converts the open problem into a specific realization problem for an operator quotient algebra.

Theorem 6.1 (New reduction: local principal quotient criterion)**.** *Let X be infinite-dimensional and let $\pi\colon \mathcal{L}(X) \to \mathcal{A} := \mathcal{L}(X)/\mathcal{SS}(X)$ be the quotient map. Suppose $K \in \mathcal{L}(X)$ satisfies*

(a) *$K \notin \Phi_+(X)$;*

(b) *every $a \in \mathcal{A}$ is either invertible or belongs to the principal left ideal $\mathcal{A}\pi(K)$.*

Then $K \in \Phi\mathcal{S}(X)$. If $\pi(K) \neq 0$, this is a strict witness $K \in \Phi\mathcal{S}(X) \setminus \mathcal{SS}(X)$.

Dually, let $\rho\colon \mathcal{L}(X) \to \mathcal{B} := \mathcal{L}(X)/\mathcal{SC}(X)$. If $K \notin \Phi_-(X)$ and every element of $\mathcal{B}$ is either invertible or lies in the principal right ideal $\rho(K)\mathcal{B}$, then $K \in \Phi\mathcal{C}(X)$, strictly so when $\rho(K) \neq 0$.

Proof. Suppose $(S, K) \in \Phi_+(X, X \oplus X)$. If $\pi(S)$ is invertible, then S is Fredholm: invertibility modulo $\mathcal{SS} \subset \mathcal{I}\backslash$ is equivalent to Fredholmness. Otherwise $\pi(S) = b\pi(K)$ for some $b \in \mathcal{A}$. Choose $B \in \mathcal{L}(X)$ representing b. Then $S - BK \in \mathcal{SS}(X)$. Therefore (BK, K) is an upper semi-Fredholm operator, because it is a strictly singular perturbation of (S, K). But

$$(BK, K) = J_B K, \qquad J_B x = (Bx, x),$$

and $J_B\colon X \to X \oplus X$ is an into isomorphism. Hence $J_B K \in \Phi_+$ if and only if $K \in \Phi_+$, contradicting (a). Thus only the first case is possible, so $S \in \Phi_+$ and $K \in \Phi\mathcal{S}$.

For the dual statement, if $[S, K] \in \Phi_-$ and $\rho(S)$ is not invertible, write $S - KB \in \mathcal{SC}$. Lower semi-Fredholm stability under strictly cosingular perturbations gives $[KB, K] \in \Phi_-$. Since

$$[KB, K] = K[B, I]$$

and $[B, I]\colon X \oplus X \to X$ is onto, $[KB, K]$ and K have the same range. This contradicts $K \notin \Phi_-$. Hence $\rho(S)$ is invertible and S is Fredholm. $\square$

Corollary 6.2 (dual-number realization target)**.** *Let $n \geq 2$.*

(i) *If there is a unital algebra isomorphism*

$$\mathcal{L}(X)/\mathcal{SS}(X) \cong \mathbb{F}[\varepsilon]/(\varepsilon^n),$$

then $\Phi\mathcal{S}(X) \neq \mathcal{SS}(X)$.

(ii) *If there is a unital algebra isomorphism*

$$\mathcal{L}(X)/\mathcal{SC}(X) \cong \mathbb{F}[\varepsilon]/(\varepsilon^n),$$

then $\Phi\mathcal{C}(X) \neq \mathcal{SC}(X)$.

Proof. Choose K representing ε. Every nonunit in the truncated polynomial algebra is divisible by ε, and K^n is strictly singular. If K were upper semi-Fredholm, then every power K^m would be upper semi-Fredholm; but a strictly singular operator on an infinite-dimensional space cannot be upper semi-Fredholm. Thus Theorem 6.1 applies. The lower semi-Fredholm/strictly cosingular argument is dual. $\square$

The criterion is stronger than merely asking for a nilpotent element modulo $\mathcal{SS}$. The known triangular example from [1, Example 2.3] already has a square-zero, non-strictly-singular operator $K(m, z) = (0, Jm)$, but it is not Φ-singular: the operator $S(m, z) = (0, z)$ has infinite-dimensional kernel while (S, K) is bounded below. In algebraic terms, the quotient contains additional nonunits not lying in the principal ideal generated by $\pi(K)$. This is exactly what condition (b) in Theorem 6.1 rules out.

7 Candidate constructions checked

The realization target in Corollary 6.2 was compared with the main "few operators" constructions.

1. On a complex hereditarily indecomposable space, every operator is a scalar plus a strictly singular operator; hence $\mathcal{L}(X)/\mathcal{SS}(X) \cong \mathbb{C}$, with no nonzero radical element [3]. Such spaces therefore cannot realize the target.
2. Tarbard's spaces have a truncated-polynomial-looking *Calkin* algebra: every operator is a polynomial in a nilpotent operator modulo the compact operators. However, the distinguished nilpotent operator is itself strictly singular [11]. Passing from compact to strictly singular operators collapses the radical and does not yield Corollary 6.2.
3. The Gowers–Maurey shift construction produces, modulo strictly singular operators, a Wiener-type algebra generated by shifts [8]. This algebra is semisimple and far from local; noninvertibility is controlled by zeros of the symbol and cannot be captured by one fixed principal radical generator.
4. The recent reflexive quotient algebra of Pelczar-Barwacz is generated by an infinite family of projections with pointwise multiplication [10]. Its many idempotents again make it nonlocal.
5. Finite direct sums or nested families of incomparable/HI spaces naturally produce diagonal or triangular quotient algebras. They supply nilpotents, but also nontrivial idempotents and the same geometric escape used in the known $P\Phi_+ \setminus \Phi\mathcal{S}$ example.

No published construction located in the search realizes a nontrivial local principal algebra as $\mathcal{L}(X)/\mathcal{SS}(X)$ or $\mathcal{L}(X)/\mathcal{SC}(X)$. This is now a sharply stated construction problem whose solution would settle the original strictness question negatively.

8 Conclusions

The supplied partial result survives detailed checking. Its four main theorems are valid, with one citation correction and one necessary qualification in the adjoint corollary. No full answer was found in the literature through 22 June 2026.

The additional deductions establish the following.

1. The global operator-ideal question is equivalent, not merely related, to the equality question $\Phi\mathcal{S} = \mathcal{SS}$ (and dually $\Phi\mathcal{C} = \mathcal{SC}$).
2. On each component, linearity would automatically imply the entire same-component multiplication stability; half of that stability holds without linearity.
3. A failure of $\Phi\mathcal{S}$ inside $P\Phi_+$ can occur only through an upper semi-Fredholm column with uncomplemented range; the dual failure requires an uncomplemented kernel.
4. A local principal quotient algebra, in particular a truncated dual-number algebra modulo strictly singular operators, would give an explicit strict counterexample.

The fourth point is the most concrete forward route uncovered here. The unresolved step is no longer an operator calculation: it is the construction (or impossibility proof) of a Banach space with the required quotient algebra. The new propositions in Sections 4–6 have not been peer reviewed and should be checked independently before citation.

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